

Chapter:- 1. Introduction

Civic issues such as potholes, garbage accumulation, streetlight failures, and water leakage are common in urban and rural areas. Often, these issues go unreported or are not resolved promptly due to a lack of an efficient communication channel between citizens and government authorities. The *Civic Issue and Resolution System* aims to bridge this gap by providing a digital platform that enables citizens to report issues, track progress, and help authorities manage and resolve complaints effectively.

Objective

- 1.To create a web-based reporting platform for easy submission of civic issues along with images and location information.
2. To develop an interactive dashboard for municipal officials to view, manage, and monitor reports by category and priority.
3. To provide automated routing and notification capabilities so that complaints are routed to the correct department and users are periodically updated.
4. To incorporate a chatbot assistant to assist users with swift responses, report filing, and real-time status checks.
5. To offer analytical insights and graphical reports to authorities to recognize repeating issues and enhance service quality.

Description

The Civic Issue and Resolution System is a web-based application designed to facilitate the reporting and resolution of public infrastructure problems. Users can login and submit complaints by specifying:

- Type of issue (e.g., roads, waste management, water, streetlights)
- Description and image
- Location (GPS or address)

Once submitted, the issue is:

- Forwarded automatically to the concerned department.
- Assigned a unique complaint ID for tracking.
- Updated through stages such as Received → In Progress → Resolved.

Authorities can log in through an admin panel to:

- View and manage all reported issues.
- Update the status.
- Generate reports based on issue type, location, and resolution time.

Technologies used may include:

- Frontend: HTML, CSS, JavaScript
- Backend: Java / Spring Boot
- Database: MySQL / PostgreSQL
- Other tools: Google Maps API, Cloud Storage for images.

Scope

The project will serve as a centralized complaint management platform for citizens and local government bodies.

In Scope:

- Citizen registration and login.
- Issue submission with image and location.
- Automated department assignment.
- Admin dashboard for authorities.
- Complaint status tracking and notifications.
- Data visualization for performance monitoring.

Out of Scope (for current phase):

- Real-time chatbot assistance.
- Automated image recognition for issue type.
- Integration with government ERP systems.

The project can later be expanded to include mobile app integration, analytics dashboards, and multilingual support.

Use Case Model**Actors:**

Citizen/User :- Reports issues and tracks their status.

Admin/Authority :- Manages, updates, and resolves issues.

Use Cases:

- Register/Login
- Report an issue
- View issue status
- Update issue status
- Generate reports

Use Case Diagram:-

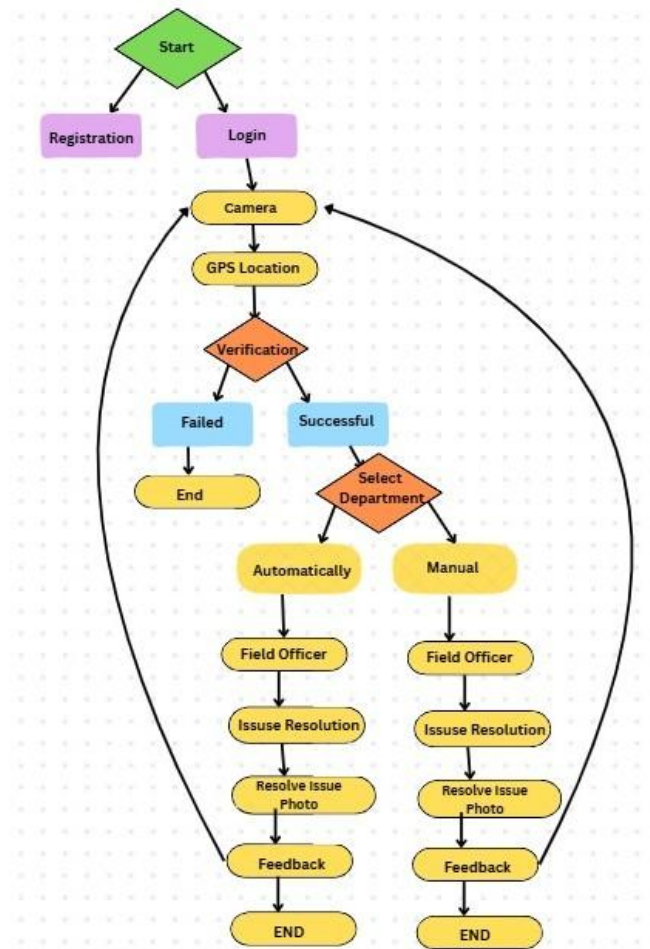


Figure:-1.Flow Chart

Chapter:-2. System Description

The Civic Issue and Resolution System is designed to provide a seamless platform for citizens to report local civic problems and for authorities to manage, track, and resolve them efficiently. This chapter provides an overview of the users, assumptions, dependencies, and both functional and non-functional requirements of the system.

Customer / User Profiles

Description:

Regular citizens or residents of a city who use the system to report issues such as potholes, garbage, water leakage, or broken streetlights.

Responsibilities:

- Register and log in to the system.
- Submit complaints with images, descriptions, and location details.
- Track complaint status and view resolution updates.
- Provide feedback or ratings once the issue is resolved.

Technical Skill Level:

Basic computer or smartphone literacy.

Administrator / Authority**Description:**

Government officials or authorized personnel responsible for managing reported issues and assigning them to the concerned departments.

Responsibilities:

- Review submitted complaints.
- Update status (e.g., Received, In Progress, Resolved).
- Communicate with citizens if additional information is needed.
- Generate analytical reports for performance review.

Technical Skill Level:

Intermediate level, comfortable using web dashboards or management systems.

Assumptions

- Users have access to the internet and a smartphone or computer.
- GPS/location services are available and enabled for accurate issue reporting.
- Each department has at least one authorized admin to monitor and resolve issues.
- Users provide accurate information and valid complaints.

Dependencies

- Internet connectivity is essential for real-time updates and communication.
- Cloud or web hosting service availability for backend operations.
- Integration with third-party services such as Google Maps API (for location tracking).
- Database dependency (e.g., MySQL or MongoDB) for data storage and retrieval.
- Browser or mobile compatibility for front-end access.

Functional Requirements :-

Table:- 1

ID	Requirement Description	Priority
1	The system shall allow users to register and log in securely.	High
2	The system shall allow users to report issues with descriptions, images, and location data.	High
3	The system shall automatically categorize issues based on type.	Medium
4	The system shall assign each issue to the appropriate department or admin.	High

5	The system shall allow users to track the status of their complaints.	High
6	The system shall allow admins to update issue status (Received, In Progress, Resolved).	High
7	The system shall send notifications or updates to users regarding issue progress.	Medium
8	The system shall allow admins to generate reports (issue count, resolution time, etc.).	Medium
9	The system shall allow users to provide feedback once an issue is resolved.	Low
10	The system shall maintain logs of all actions for accountability.	Medium

Non-Functional Requirements:-

Table:-2

Category	Requirement Description
Performance	The system should handle multiple concurrent users efficiently with minimal response time.
Scalability	The system should support future expansion to multiple cities or departments.
Security	All user data and credentials must be encrypted. Access control should be role-based.
Reliability	The system should ensure high uptime (>99%) with backup and recovery mechanisms.
Usability	The user interface should be intuitive, responsive, and mobile-friendly.
Maintainability	The system should be modular, allowing for easy updates and bug fixes.
Availability	The service should be available 24/7 for reporting and tracking issues.

Compatibility	The system should work across major browsers and devices (mobile and desktop).
Data Integrity	All submitted complaints and updates must be stored consistently and without data loss.
Localization (Optional)	The system can support multiple languages for diverse user accessibility.

Chapter:-3. Design

This chapter describes the overall design of the Civic Issue and Resolution System. It includes system architecture, Entity–Relationship (E-R) model, Data Flow Diagrams (DFDs), and database schema. The design ensures smooth flow of information, efficient data management, and clarity in system interactions between users, administrators, and the application backend.

System Design

The system is designed using a three-tier architecture:

Presentation Layer (Frontend):

Provides an intuitive interface for users and administrators to interact with the system.

Technologies: HTML, CSS, JavaScript

Application Layer (Backend):

Handles business logic, authentication, complaint routing, and communication between the frontend and the database.

Technologies: Java/ Spring Boot

Data Layer (Database):

Stores all persistent data such as user details, complaints, department information, and complaint statuses.

Technologies: MySQL / PostgreSQL

E-R Diagram

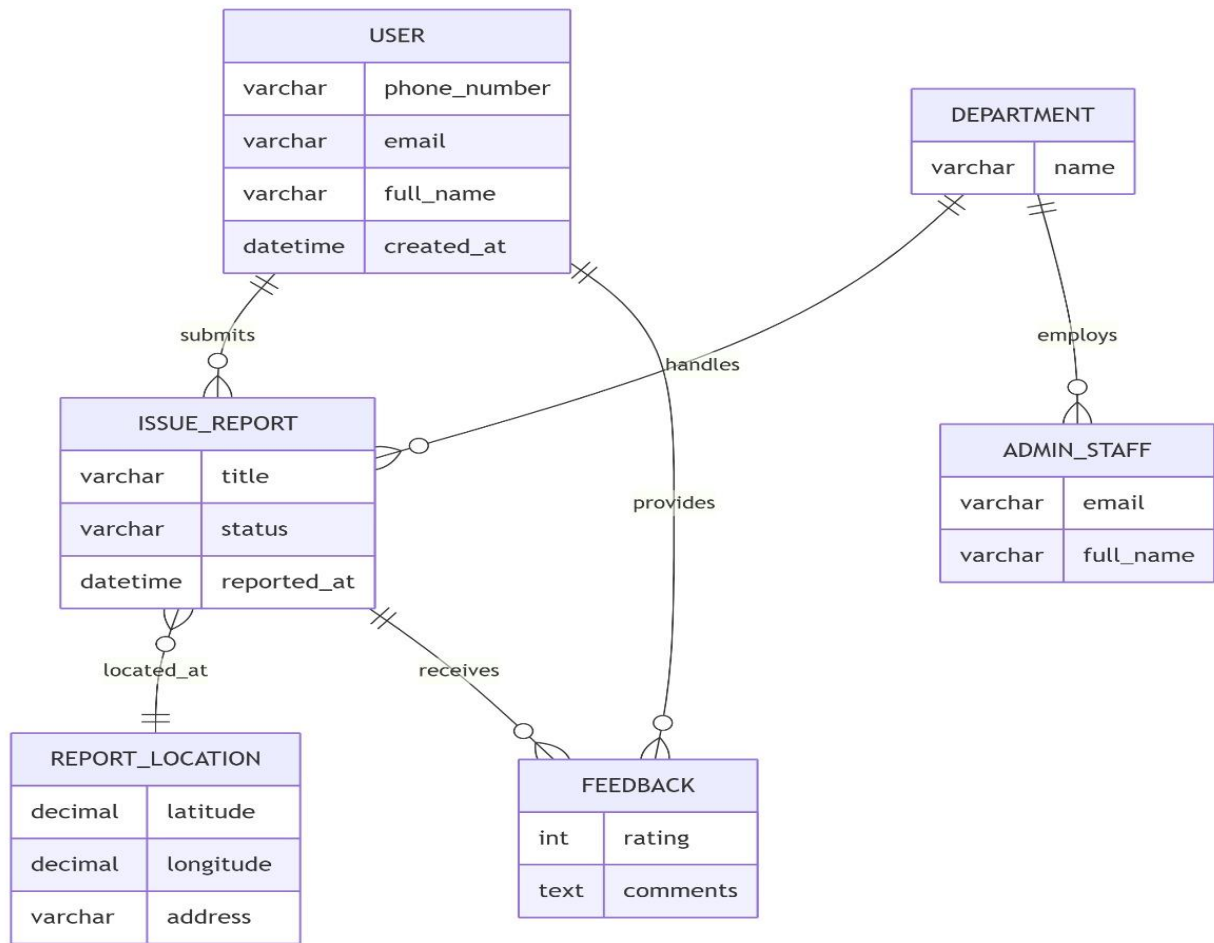


Figure:-2

DFD'S:-

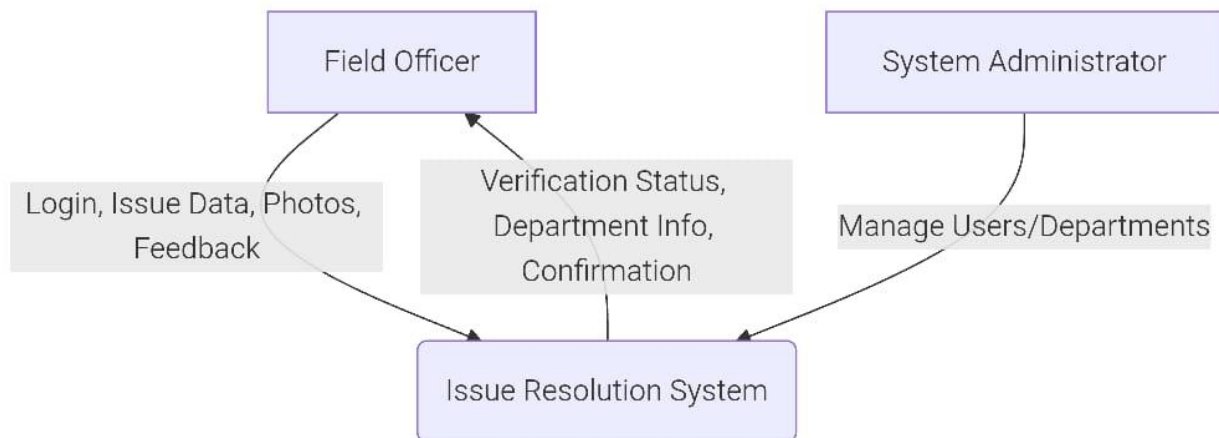


Figure:-3

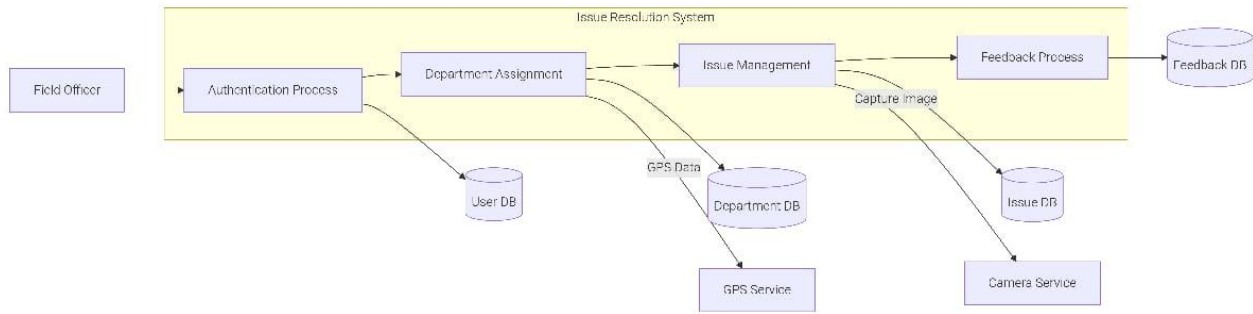


Figure:-4

Database Design:-

The database design ensures structured storage and fast retrieval of information. Below is a sample relational schema for the Civic Issue and Resolution System.

Table:3

Table Name	Description
users	Stores user login and personal details.
departments	Stores details of civic departments.
complaints	Stores all user complaints.
admins	Stores admin login credentials and their assigned departments.
feedback	Stores feedback given by users.

Table:- 4

Users Entity

Field Name	Data Type	Description
name	VARCHAR(100)	User's full name
email	VARCHAR(100)	User's email
password	VARCHAR(255)	Encrypted password
phone	VARCHAR(15)	Contact number
address	VARCHAR(255)	User address

Table:- 5

Departments Entity

Field Name	Data Type	Description
dept_id	INT (PK)	Department ID
dept_name	VARCHAR(100)	Department Name
dept_email	VARCHAR(100)	Department contact email

Table:-6

Admins Entity

Field Name	Data Type	Description
admin_id	INT (PK)	Admin ID
admin_name	VARCHAR(100)	Name of the admin
dept_id	INT (FK)	Assigned department
email	VARCHAR(100)	Admin email
password	VARCHAR(255)	Encrypted password

Table:-7

Complaints Entity

Field Name	Data Type	Description
complaint_id	INT (PK)	Unique complaint ID
user_id	INT (FK)	User who raised the complaint
dept_id	INT (FK)	Department handling the issue
title	VARCHAR(255)	Short title of complaint

description	TEXT	Detailed description
image_url	VARCHAR(255)	Uploaded image link
location	VARCHAR(255)	Complaint location
status	VARCHAR(50)	Complaint status (Pending/In Progress/Resolved)
created_at	DATETIME	Complaint creation date
updated_at	DATETIME	Last update time

Table:-8

Feedback Entity

Field Name	Data Type	Description
feedback_id	INT (PK)	Feedback ID
complaint_id	INT (FK)	Related complaint
user_id	INT (FK)	Feedback giver
rating	INT	Rating (1–5)
comments	TEXT	User comments
feedback_date	DATETIME	Date of feedback submission

Chapter: 4.Scheduling and Estimates

The development of the Civic Issue and Resolution System was planned and executed over a period of five weeks. This schedule was designed to ensure systematic research, analysis, and development of both frontend and backend modules while establishing proper database connectivity and user authentication functionalities.

Project Schedule Overview

The project followed a phased development approach, ensuring that each stage built upon the previous one. The five-week schedule is summarized below.

Table:-9

Week	Phase / Activity	Description of Work Done	Deliverables / Outcome
Week 1	Project Research and Selection	Identified common civic issues and existing systems. - Selected “Civic Issue and Resolution System” as project topic. - Conducted literature survey and feasibility study. - Defined objectives, problem statement, and scope.	Project proposal and finalized system requirements.
Week 2	System Analysis and Design	Prepared functional and non-functional requirements. - Designed E-R diagram, DFDs, and database schema. - Selected appropriate tech stack (HTML/CSS/JS + Python/Flask + MySQL).	Completed system design and database schema.
Week 3	Backend Development	Set up project structure and environment. - Implemented backend logic using Flask/Django. - Created database tables (users, complaints, departments). - Configured routes for registration and login.	Functional backend with working database connection.
Week 4	Frontend Development and Connectivity	Developed user interface for login, registration, and home page using HTML, CSS, and JS. - Integrated frontend with backend APIs. - Implemented form validation and error handling.	Connected frontend-backend with live SQL data exchange.
Week 5	Integration and Testing	Connected all modules (login → home page flow). - Tested user registration, login, and dashboard navigation. - Fixed bugs and optimized performance.	Working prototype up to login and home page, tested successfully.

Table:-10

Effort and Time Estimation

Activity	Estimated Time (Hours)	Actual Time (Hours)	Remarks
Project Research & Requirement Gathering	10	12	Included documentation and topic selection.
System Design & Database Schema	12	14	Covered E-R and DFD diagrams.
Backend Development (Flask + SQL)	16	18	Included user registration and login API setup.
Frontend Development (UI + Forms)	14	16	Included login, registration, and home page design.
Integration & Testing	8	10	Included end-to-end connectivity and debugging.

Total Estimated Hours	60 hours	70 hours	Slight increase due to testing and connectivity refinements.
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Tools and Technologies Use

Table:-11

Category	Tools / Technologies
Frontend	HTML5, CSS3, JavaScript
Backend	Python (Flask/Django)
Database	MySQL
Development Tools	VS Code / PyCharm
Connectivity	SQLAlchemy / MySQL Connector
Testing	Manual testing via browser and Postman
Documentation	MS Word, Draw.io (for diagrams)

At the end of 5 weeks, a functional prototype of the *Civic Issue and Resolution System* was achieved, including:

- Working login and registration module
- Backend–frontend connectivity with database
- SQL connection with live data retrieval
- Navigable home page dashboard.